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# Covariate Shift Regularization with Kernels

A  
**STUDENT PROJECT**  
report

by

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# Abstract

Lorem Ipsum

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# 1 Introduction

Lorem Ipsum

## 2 Overview: Covariate Shift

Covariate shift occurs when the marginal distribution of the input variables  $\mathbb{P}_{\text{train}}(\mathcal{X})$  differs between the training and testing phases, while the conditional distribution of the output given the input  $\mathbb{P}(\mathcal{Y}|\mathcal{X})$  remains unchanged.

Mathematically, this is expressed as:

$$\begin{aligned}\mathbb{P}_{\text{train}}(\mathcal{X}) &\neq \mathbb{P}_{\text{test}}(\mathcal{X}) \\ \mathbb{P}_{\text{train}}(\mathcal{Y}|\mathcal{X}) &= \mathbb{P}_{\text{test}}(\mathcal{Y}|\mathcal{X})\end{aligned}$$

The goal is to learn a function  $f$  during training, based on data  $(X_i, Y_i)_{i=1}^n \sim \mathbb{P}_{\text{train}}(\mathcal{X}, \mathcal{Y})$ , that generalizes well on the test distribution  $\mathbb{P}_{\text{test}}(\mathcal{X}, \mathcal{Y})$ . The change is to adjust for the mismatch between  $\mathbb{P}_{\text{train}}(\mathcal{X})$  and  $\mathbb{P}_{\text{test}}(\mathcal{X})$ , which can degrade model performance.

## 3 Overview: Kernel Methods

Kernel methods operate by embedding input data  $X \in \mathcal{X}$  into a high-dimensional feature space  $\mathcal{H}$  through a mapping function  $\phi : \mathcal{X} \rightarrow \mathcal{H}$ . Instead of explicitly computing the mapping  $\phi(X)$ , kernel methods compute inner products in the feature space using a **kernel function**  $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ , where:

$$k(X, X') = \langle \phi(X), \phi(X') \rangle_{\mathcal{H}}$$

Common kernel functions include:

- **Linear kernel:**  $k(X, X') = X^\top X'$
- **Radial Basis Function (RBF) kernel:**  $k(X, X') = \exp(-\gamma \|X - X'\|^2)$
- **Polynomial kernel:**  $k(X, X') = (X^\top X' + c)^d$

### 3.1 Covariate Shift in Kernel Methods

In the kernel framework, the objective is to find a function  $f$  in the reproducing Kernel Hilbert Space (RKHS)  $\mathcal{H}$  induced by the kernel function  $k(\cdot, \cdot)$ . Given training data  $(X_i, Y_i)_{i=1}^n$ , we aim to minimize the risk:

$$\mathcal{R}(f) = \mathbb{E}_{\mathbb{P}_{\text{train}}}[\ell(Y, f(X))]$$

where,  $\ell(\cdot, \cdot)$  is a loss function (e.g., squared error for regression, cross-entropy for classification). However, under covariate shift the risk we care about is with respect to the test distribution  $\mathbb{P}_{\text{test}}$ :

$$\mathcal{R}_{\text{test}}(f) = \mathbb{E}_{\mathbb{P}_{\text{test}}}[\ell(Y, f(X))]$$

Since  $\mathbb{P}_{\text{test}}(X) \neq \mathbb{P}_{\text{train}}(X)$ , the standard empirical risk minimization (ERM) on training data may not perform well on the test data.

**Empirical Risk Minimization**

We recall the goal of supervised learning: Given some observations  $(x_j, y_j) \in \mathcal{X} \cdot \mathcal{Y}$  ( $j = 1, 2, \dots, n$ ) of inputs and outputs (or features/variables) that we call *training data* and given a new  $x \in \mathcal{X} = \mathbb{R}^d$ , predict  $y \in \mathcal{Y} = \mathbb{R}$  (*test data*) with a regression function  $f : \mathcal{X} \rightarrow \mathcal{Y}$  such that  $f(x) \approx y$ .

Given data  $(x_j, y_j) \in \mathcal{X} \times \mathcal{Y}$ ,  $j = 1, 2, \dots, n$ , the empirical risk is defined by:

$$\hat{\mathcal{R}}(\theta) := \frac{1}{n} \sum_{j=1}^n (y_j - f_\theta(x_j))^2, \theta \in \Theta$$

To obtain a "good" or "reasonable" estimator, we minimize the empirical risk.

**3.1.1 Adjusting for Covariate Shift**

To account for the shift in the distributions, we introduce **importance weighting**. The idea is to reweight the training data to match the test distribution. If the **Radon-Nikodym derivative**  $w(X) = \frac{d\mathbb{P}_{\text{test}}(X)}{d\mathbb{P}_{\text{train}}(X)}$  (i.e., the ratio of the test distribution to the training distribution) is known or can be estimated, we can minimize the weighted empirical risk:

$$\mathcal{R}_{\text{weighted}}(f) := \frac{1}{n} \sum_{i=1}^n w(X_i) \ell(Y_i, f(X_i))$$

This ensures that the training data is effectively "corrected" to resemble the test distribution.

**3.1.2 Kernel Ridge Regression under Covariate Shift**

Let us consider **Kernel Ridge Regression (KRR)**, a kernel-based method for regression that minimizes a regularized empirical loss (with the *least square loss*). Without covariate shift, the solution  $f \in \mathcal{H}$  is found by solving:

$$\hat{f} = \arg \min_{f \in \mathcal{H}} \left\{ \frac{1}{n} \sum_{i=1}^n (Y_i - f(X_i))^2 + \lambda \|f\|_{\mathcal{H}}^2 \right\}, \quad (3.1)$$

where  $\lambda > 0$  is a regularization parameter that controls overfitting.

To adjust for the covariate shift, we apply the importance weights to the loss function:

$$\hat{f} = \arg \min_{f \in \mathcal{H}} \left\{ \frac{1}{n} \sum_{i=1}^n w(X_i) (Y_i - f(X_i))^2 + \lambda \|f\|_{\mathcal{H}}^2 \right\} \quad (3.2)$$

The solution to this problem can be written in terms of the kernel matrix  $K \in \mathbb{R}^{n \times n}$  with entries  $K_{ij} = k(X_i, X_j)$ , and the importance weights can be incorporated into a diagonal

matrix  $W = \text{diag}(w(X_1), \dots, w(X_n))$ . The closed-form solution for the KRR estimator under covariate shift is given by:

$$\hat{f} = (KWK + \lambda I)^{-1}KWY, \quad (3.3)$$

where  $Y = (Y_1, \dots, Y_n)^\top$  is the vector of training labels. This solution adjusts for the shift by weighting the kernel matrix and labels according to the test distribution.

### 3.1.3 Estimating Importance Weights

In practice, the true density ratio  $w(X) = \frac{\mathbb{P}_{\text{test}}(X)}{\mathbb{P}_{\text{train}}(X)}$  is often unknown. There are several strategies to estimate  $w(X)$ , including:

- **Kernel Density Estimation (KDE)**: Estimate both  $\mathbb{P}_{\text{train}}(X)$  and  $\mathbb{P}_{\text{test}}(X)$  using kernel density estimators and compute the ratio.
- **Logistic Regression-based Density Ratio Estimation**: Train a logistic regression model to distinguish between samples from the training and test distributions. The estimated probabilities can be used to approximate the density ratio.
- **KLIEP (Kullback-Leibler Importance Estimation Procedure)**: Directly estimate the ratio by minimizing the Kullback-Leibler divergence between the reweighted training distribution and the test distribution.

Once  $w(X)$  is estimated, it can be plugged into the weighted KRR formulation to adjust for the covariate shift.

### 3.1.4 Self-Attention for Kernel Methods under Covariate Shift

Another approach is the **self-attention mechanism**, where the importance weights  $w(X)$  are learned dynamically along with the model parameters. This can be formulated as a joint optimization problem, where the objective is to **minimize the empirical loss with respect to both the regression function  $f$  and the weights  $w(X)$** . The optimization alternates between updating the model parameters  $f$  and the weights  $w(X)$  using gradient-based methods.

In kernel-based methods, this can be implemented by introducing learnable parameters  $w_i$  for each sample, and optimizing both the KRR objective and the weights simultaneously:

$$\min_{f, w} \left\{ \frac{1}{n} \sum_{i=1}^n w(X_i) \ell(Y_i, f(X_i)) + \lambda \|f\|_{\mathcal{H}}^2 \right\} \quad (3.4)$$

subject to constraint on  $w(X)$  (e.g., positivity, normalization). This allows the model to adapt the importance weights based on the data characteristics, potentially improving performance under complex covariate shift scenarios.

# Bibliography